



PGCView_v2 Image Annotation Project Report

Project Name: PGCView_v2 Image Annotation Project

Client: Bo Meyering

Annotation Team: Shayan Khan, Muhammad Haris

Tool Used: Labelbox

Date: [Insert Date]

1. Project Overview

The project required annotating images using **two annotation types**: bounding boxes (for regions of interest) and segmentation masks (for per-pixel categories). The client uploaded the dataset, and our team completed the annotations in accordance with the provided guidelines.

2. Dataset Description

- **Total Images:**[Insert Number of Images]
- **Total classes:** 14
- **Resolution:** 4032 × 3024
- **Source:** Uploaded by the Land-Institute
- **Annotation Types:** Bounding Boxes and Segmentation Masks

1. Bounding Box Classes

Bounding boxes identified regions of interest (ROI) within agricultural plots:

- **quadrat_bb**: Bounding box for the corner elbows of a PVC (polyvinyl chloride / Plastic Pipe) sampling quadrat.
- **marker_bb**: Bounding box for the colored markers.

Each image contained **only one type of ROI marker** – either colored markers or a PVC quadrat, never both.

2. Segmentation Classes

Segmentation masks were created for pixel-level categorization:

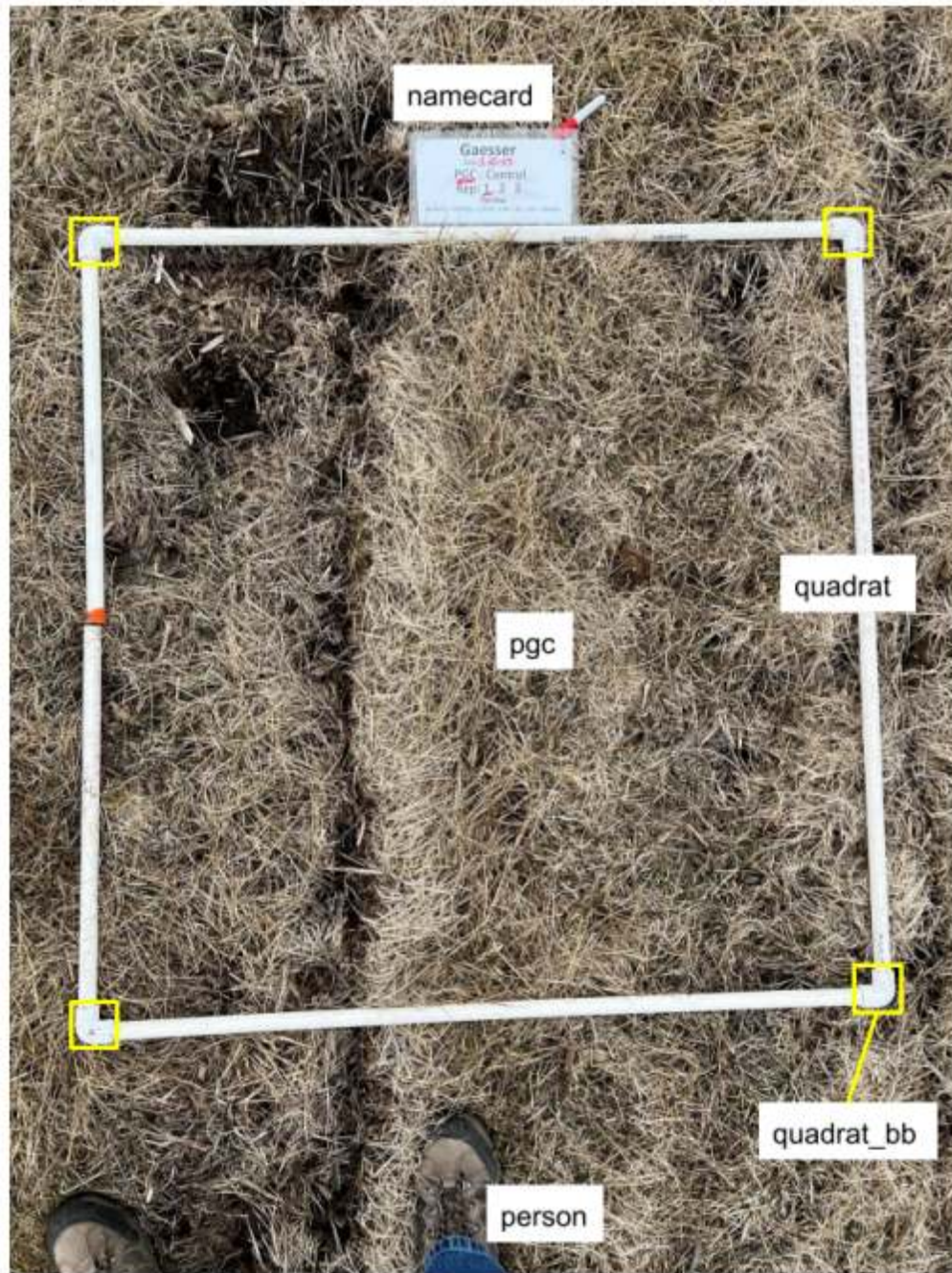
- **soil**: Dirt, rocks (background class).
- **Stover**: Crop residue such as corn stalks, dried corn leaves, sticks.
- **pgc**: Perennial Groundcover – planted grass and weedy grass.
- **weed**: Broadleaf weeds (excluding weedy grass).
- **corn**: Living corn leaves and stalks.
- **soybean**: Living soybean plants and stalks.
- **marker**: Colored markers (blue/red).
- **quadrat**: PVC sampling quadrat mask.
- **snow**: Snow or ice (**rare class**).
- **namecard**: White placard with text/plot ID (**rare class**).
- **person**: Shoes, legs, or hands of the photographer (**rare class**).
- **marking_flag**: Colored marking flag/stake (**rare class**).

Note: Complex soil regions were sometimes left partially unlabeled. During model training, unlabeled pixels were aggregated into the soil class.

Examples



[Figure 2.1]



[Figure 2.2]

3. Annotation Tasks in Labelbox

- **Tool Used:** Labelbox (cloud-based annotation tool)
- **Annotation Types:** Bounding Boxes and Segmentation Masks
- **Workflow in Labelbox:**

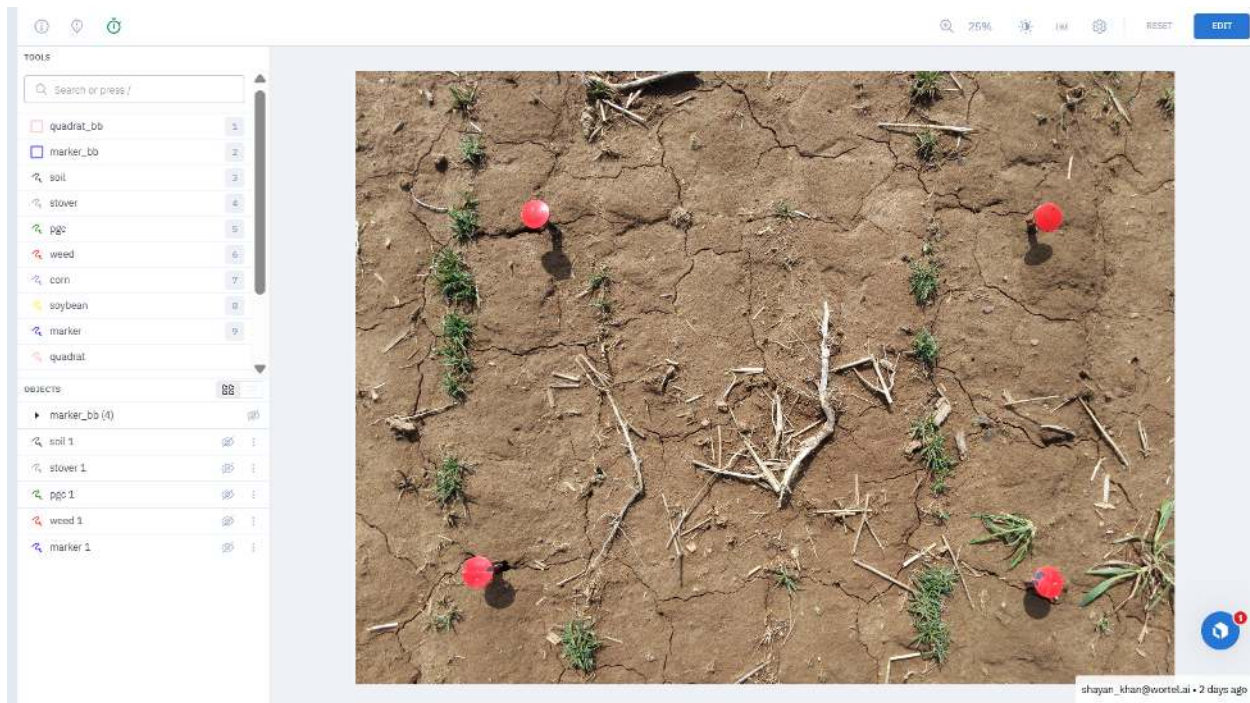
1. Project created with predefined classes shared by the client.
2. The client uploaded the dataset into Labelbox.
3. Annotated images manually using the Labelbox interface.
4. Using model-assisted labeling[auto label] to reduce manual work.
5. If there were any issues, the client rejected the annotation. We fixed and resubmitted the images until approval.

4. Quality Assurance

1. All annotations were **reviewed by the client** to ensure correctness.
2. If issues were found, the client **rejected the image**, and we corrected and resubmitted it.
3. Verified annotations using **Labelbox's built-in review panel**.
4. Corrected **overlapping or inaccurate bounding boxes**.
5. Ensured **consistent labeling** across both target and rare classes.

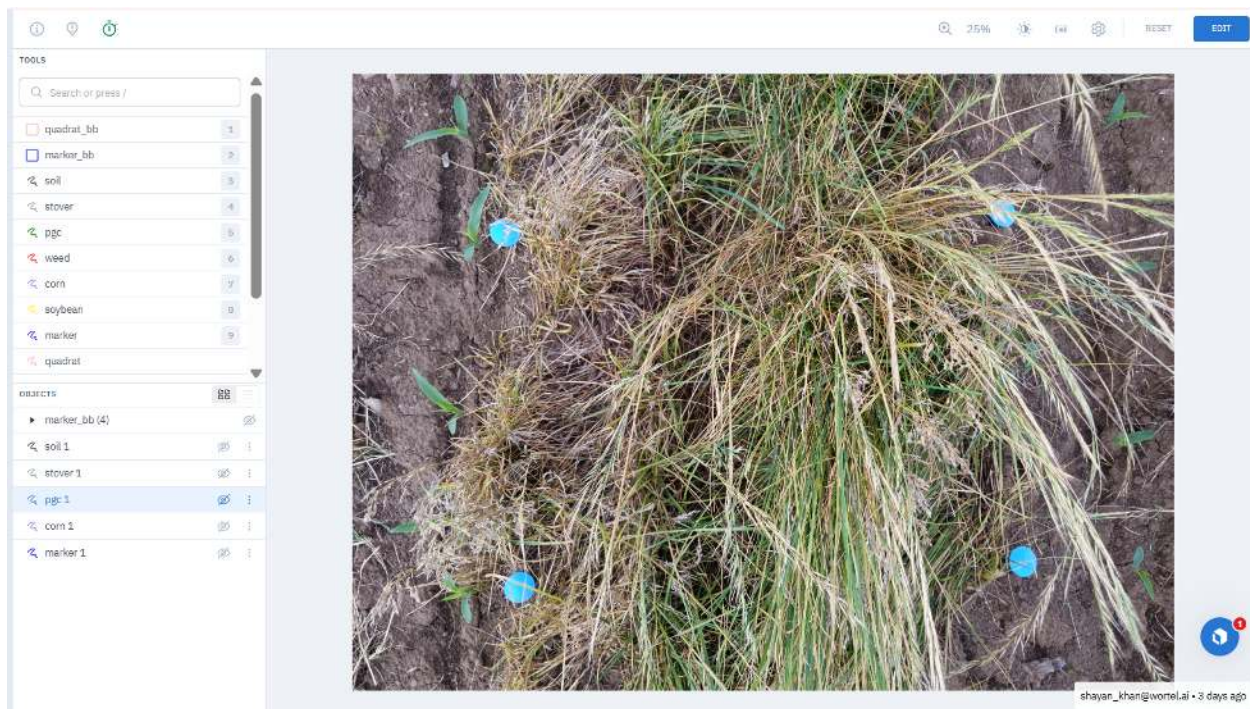
5. Final Output

- **Total Images Annotated:** 2275
- **Total Objects Labeled:** 16,647
- **Annotation Types:** Bounding Boxes + Segmentation Masks
- **Example Screenshots:**

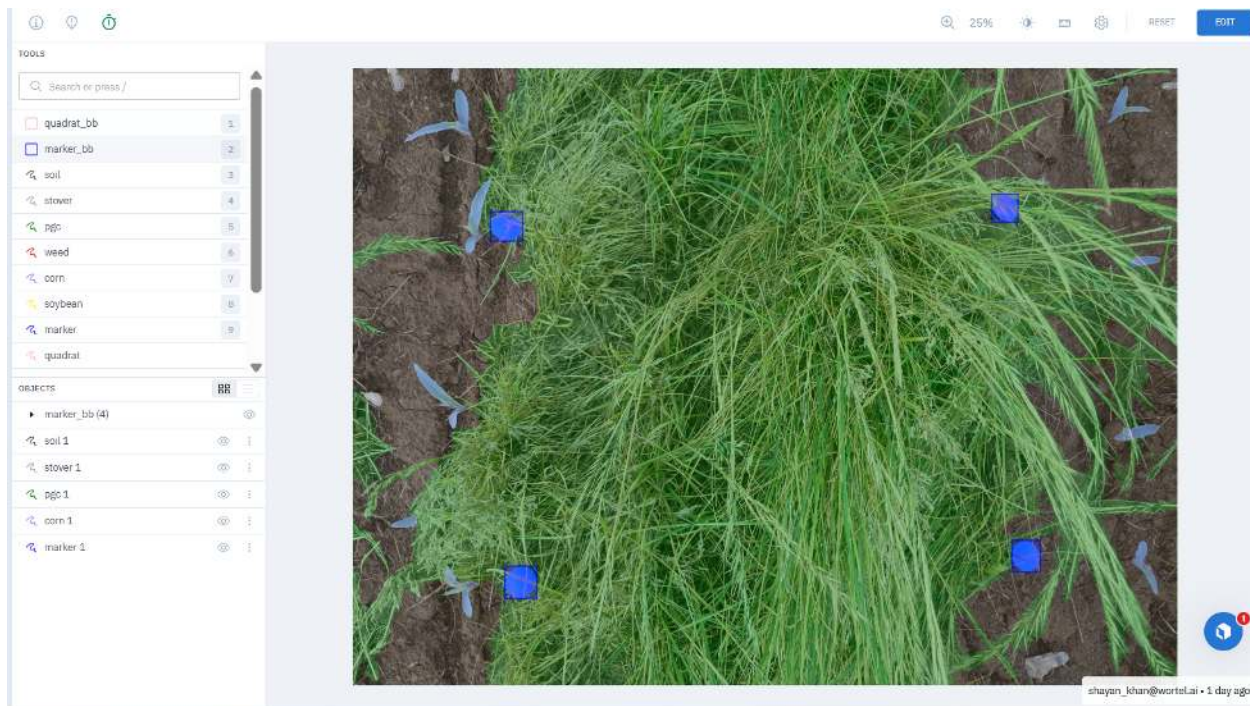


[Figure 5.1] Before Annotations.





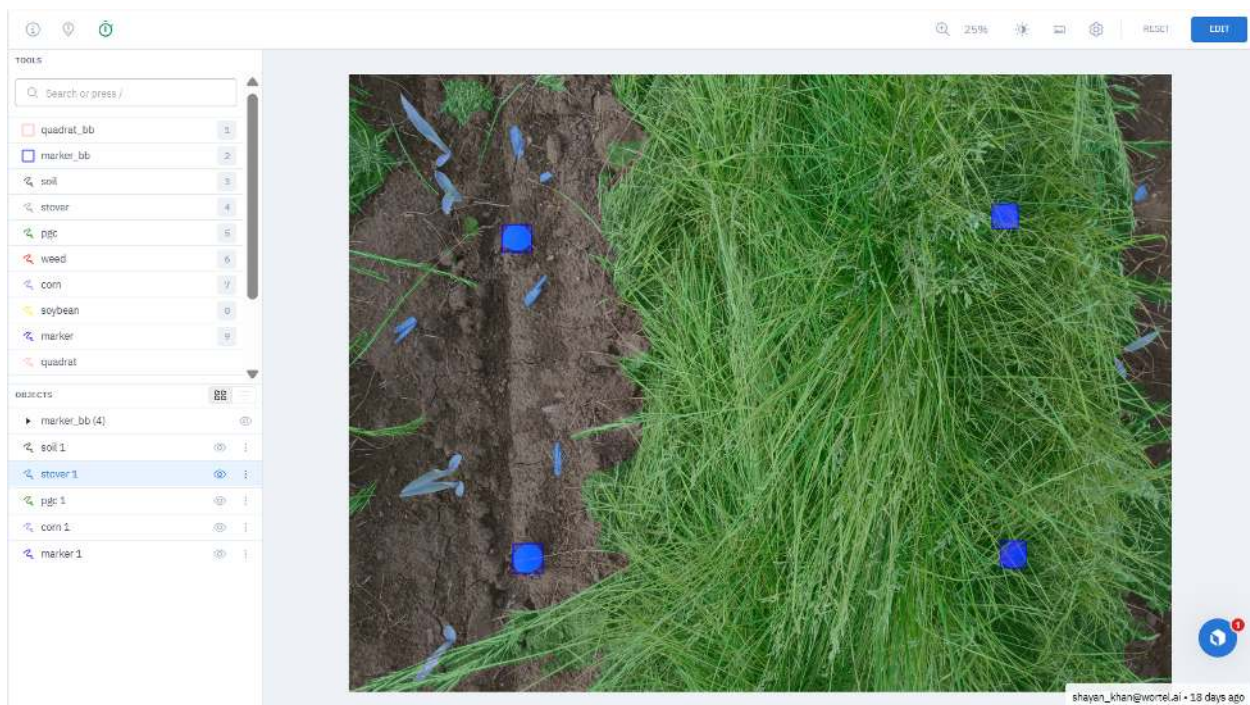
[Figure 5.2] Before Annotations.



[Figure 5.2] After annotation. Annotated objects in this image include: soil, stover, pgc, corn, marker, and marker_bb.



[Figure 5.3] Before Annotations.

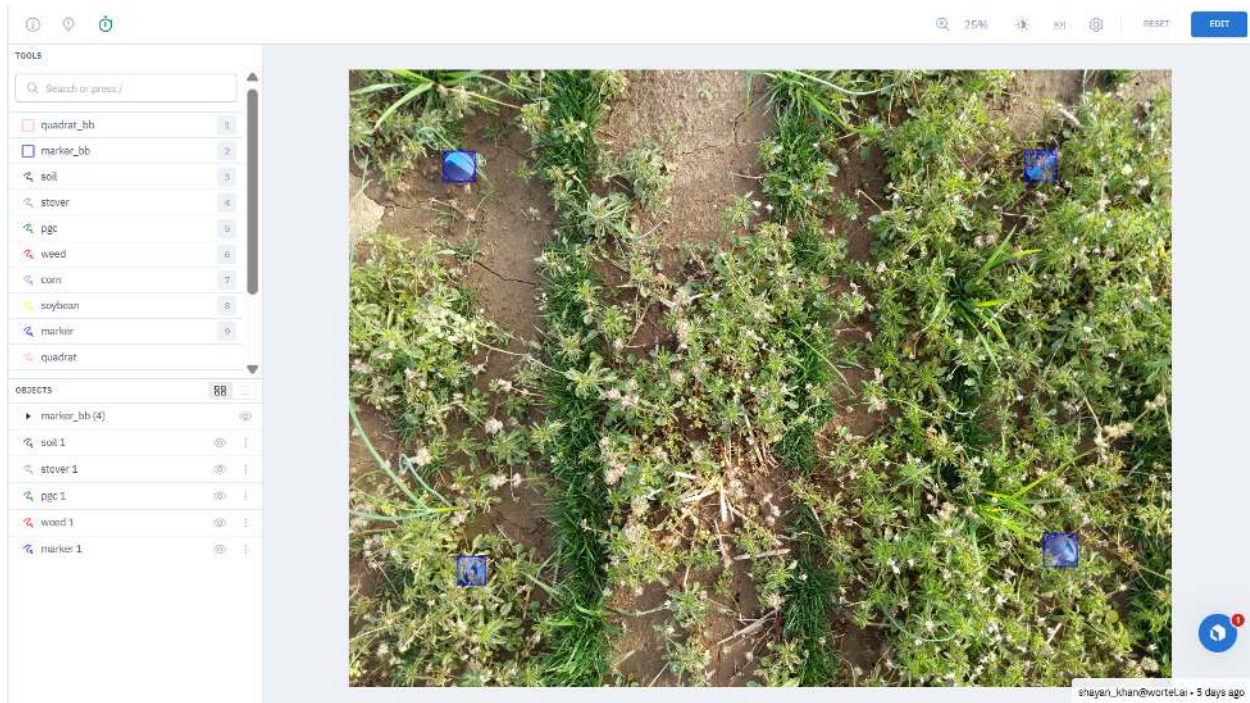




[Figure 5.4] Before annotation.



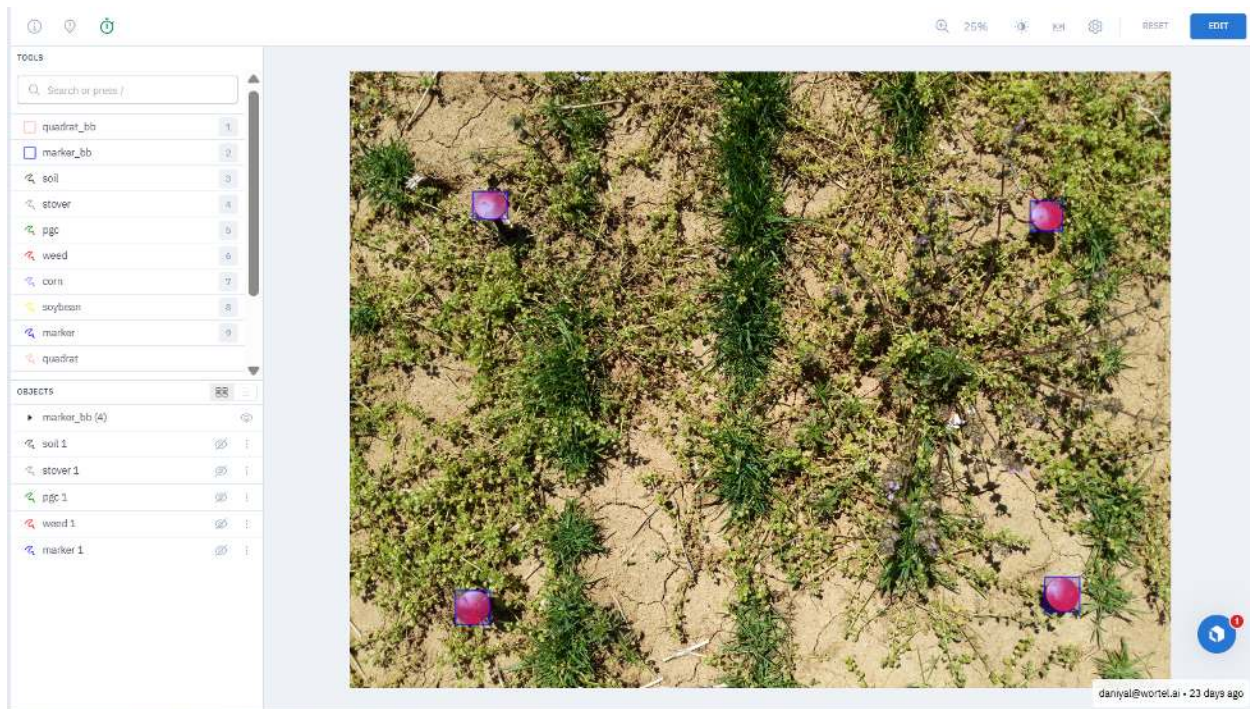
[Figure 5.4] After annotation. Annotated objects in this image include: soil, stover, pgc, weed, quadrat, quardrat_bb, and person.



[Figure 5.5] Before annotation.



[Figure 5.5] Image accepted by the client.



[Figure 5.6] Before Annotations.

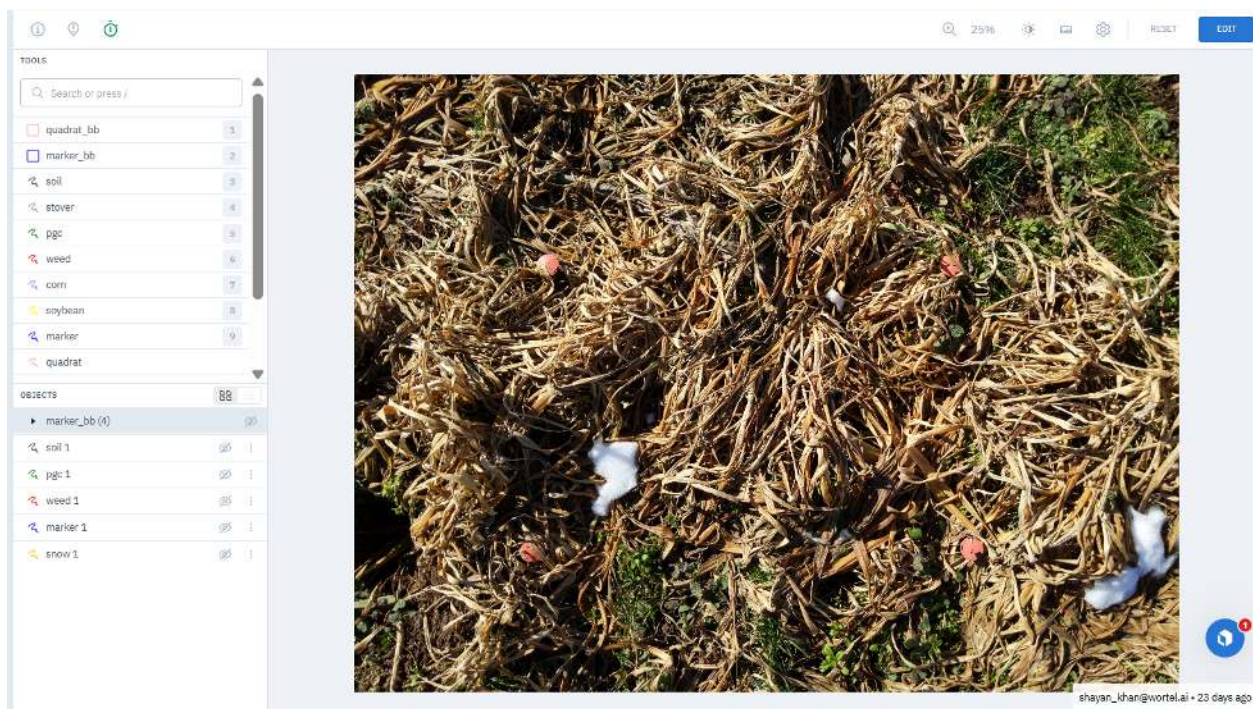




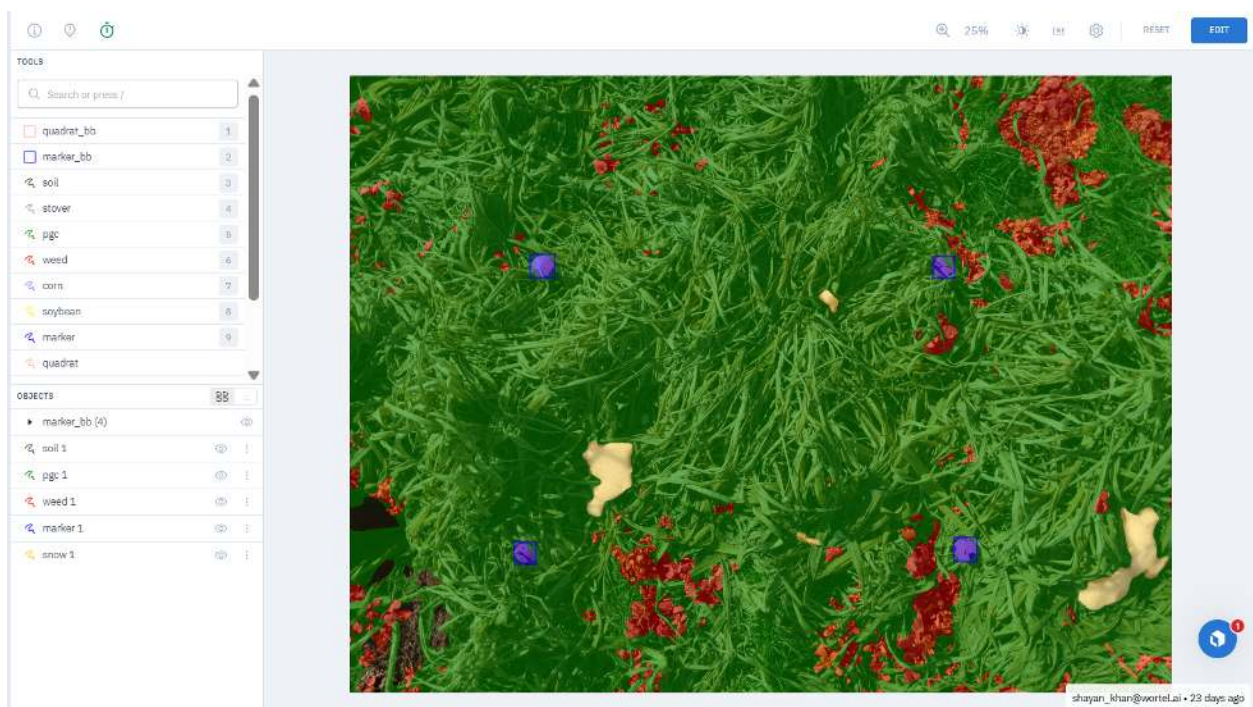
[Figure 5.7] Before Annotations.

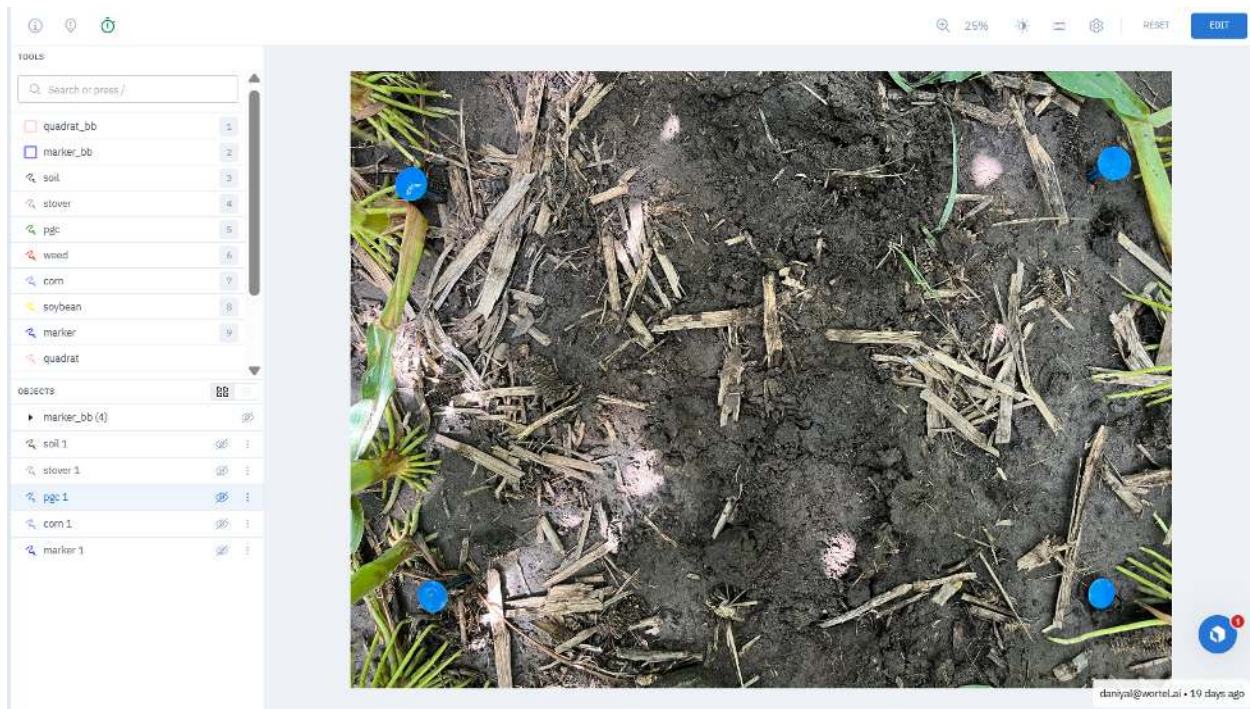






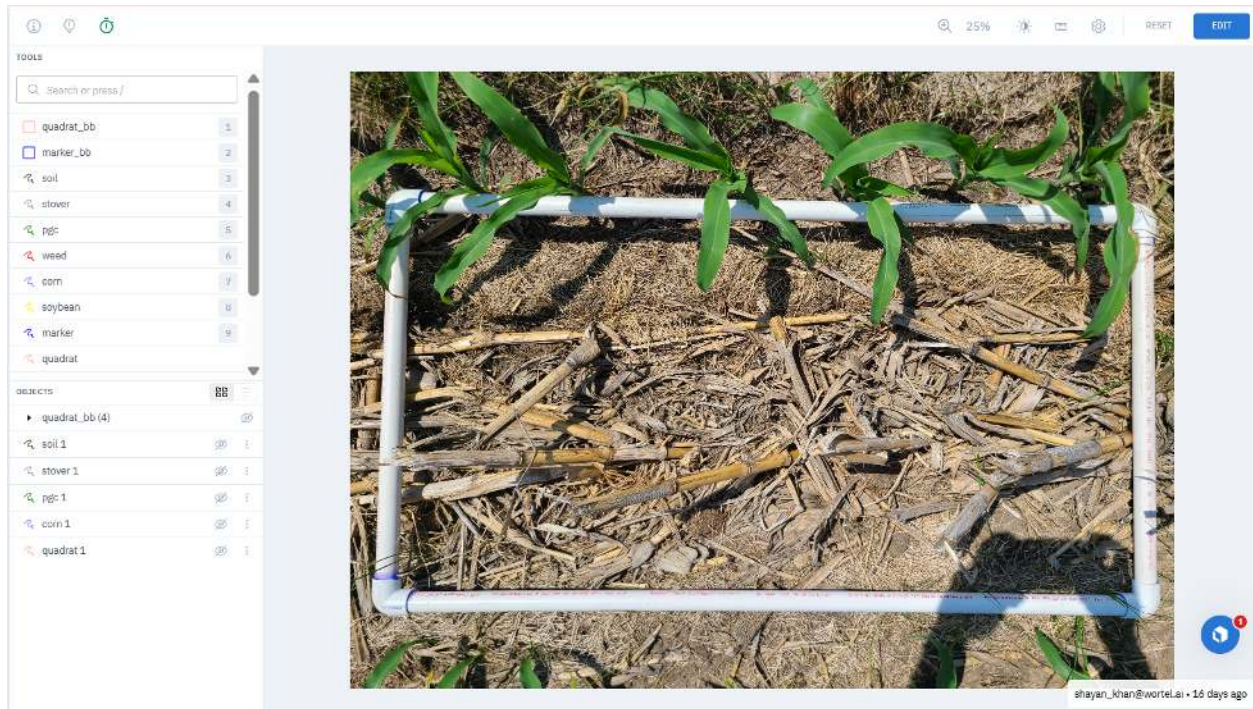
[Figure 5.9] Before Annotations.





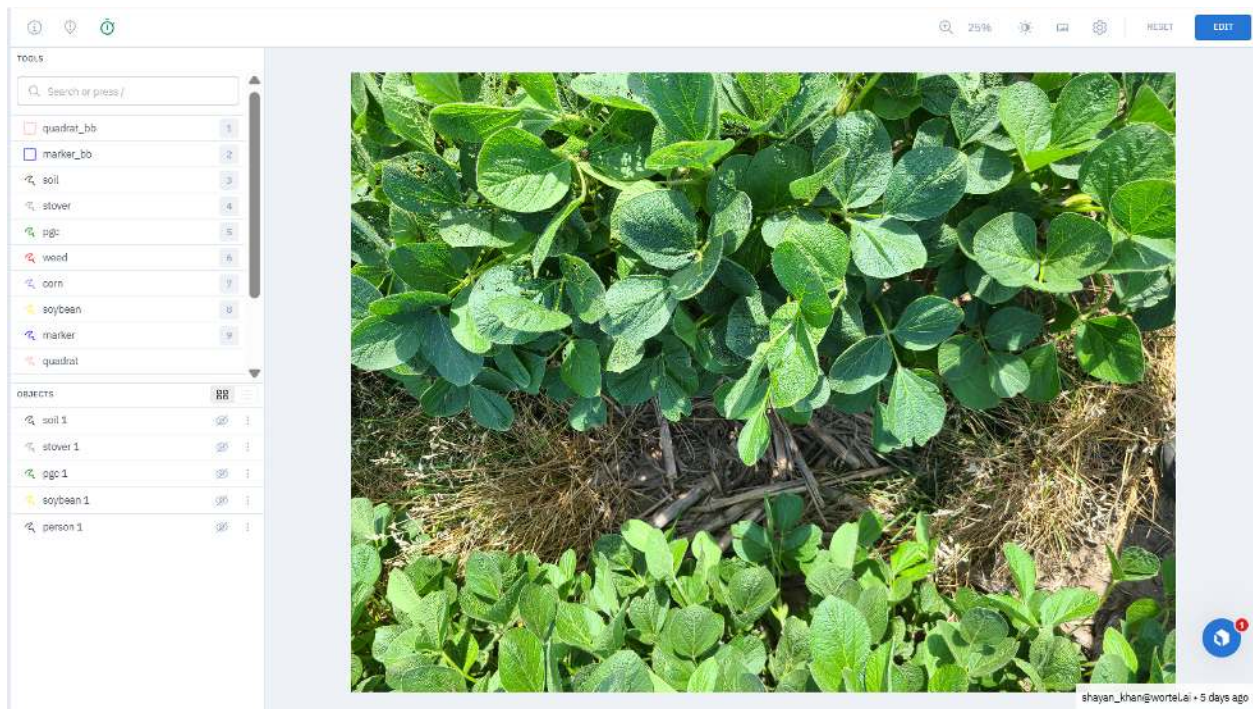
5.10



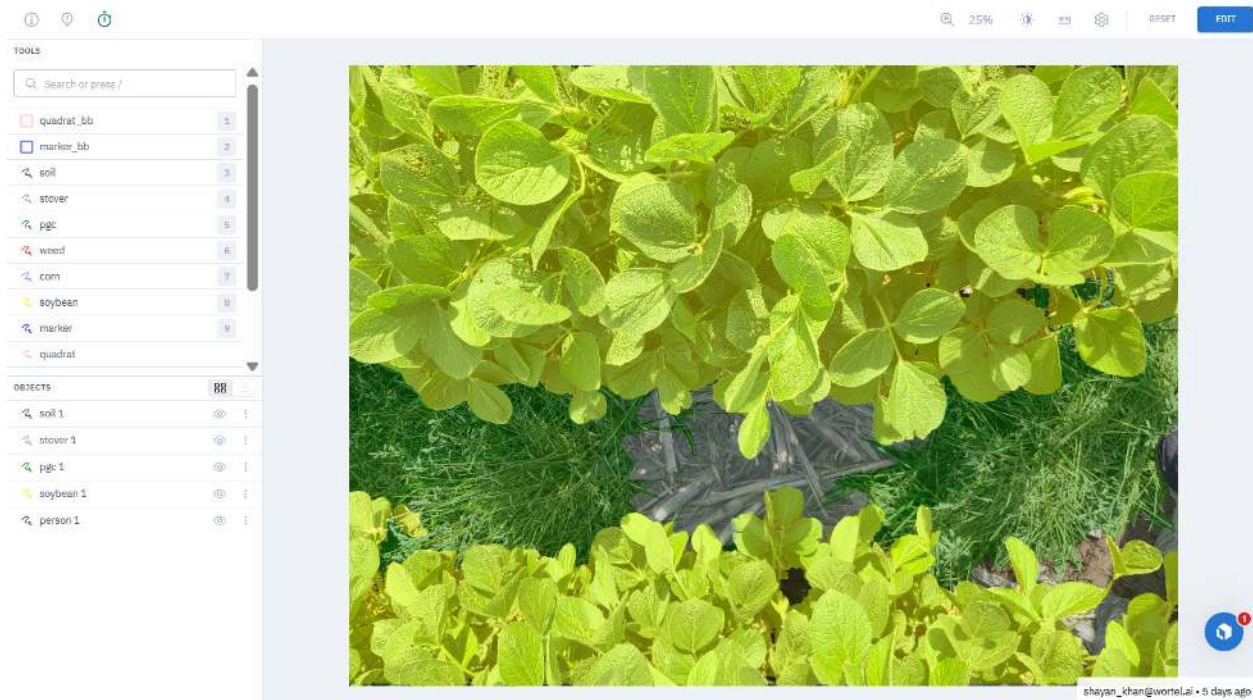


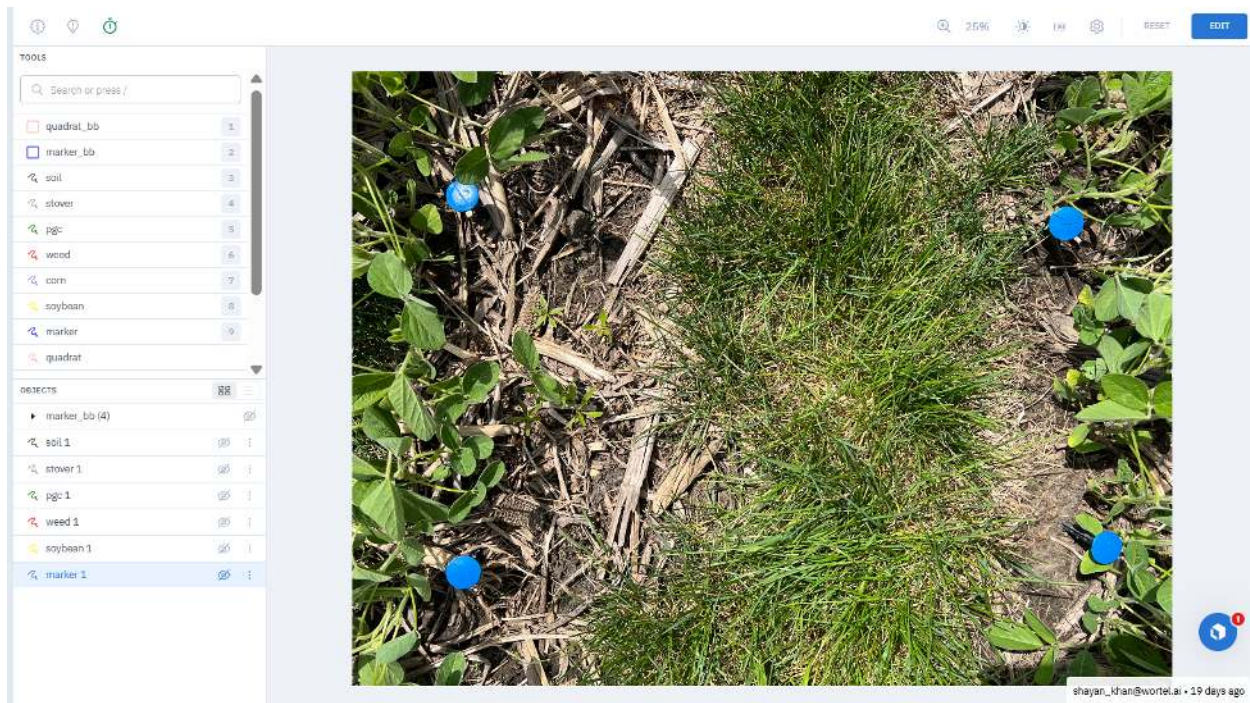
[Figure 5.11] Before Annotations.



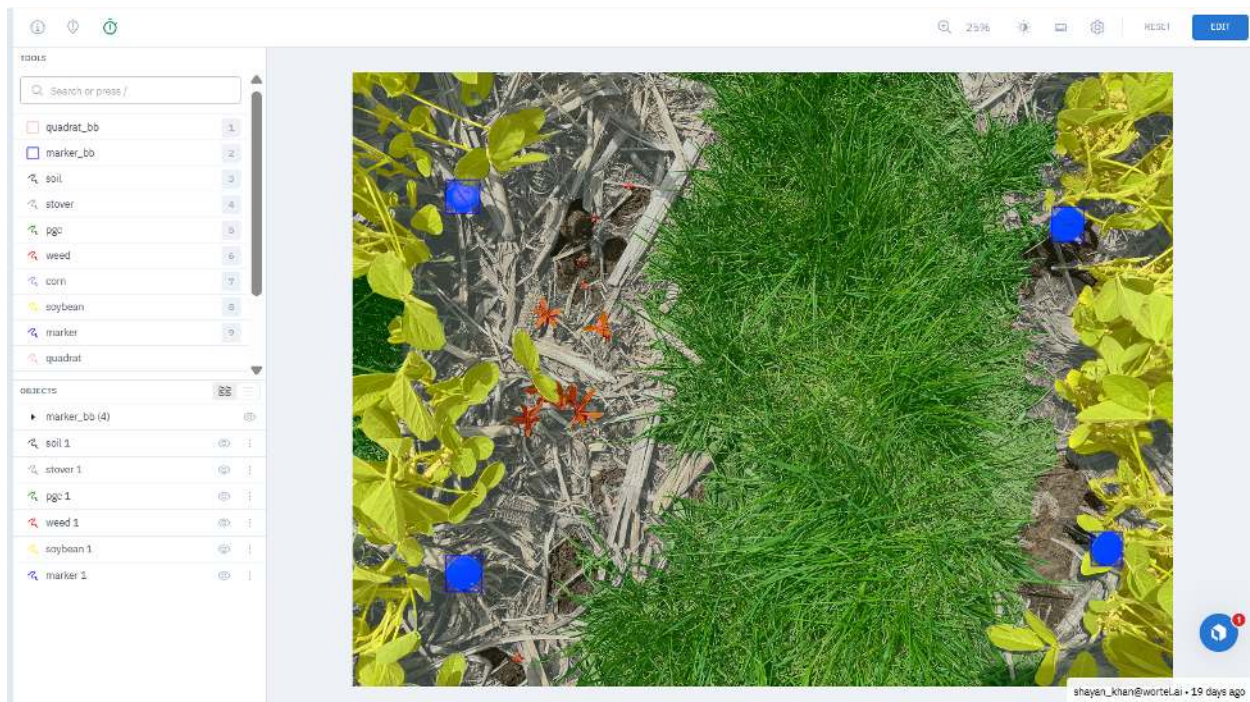


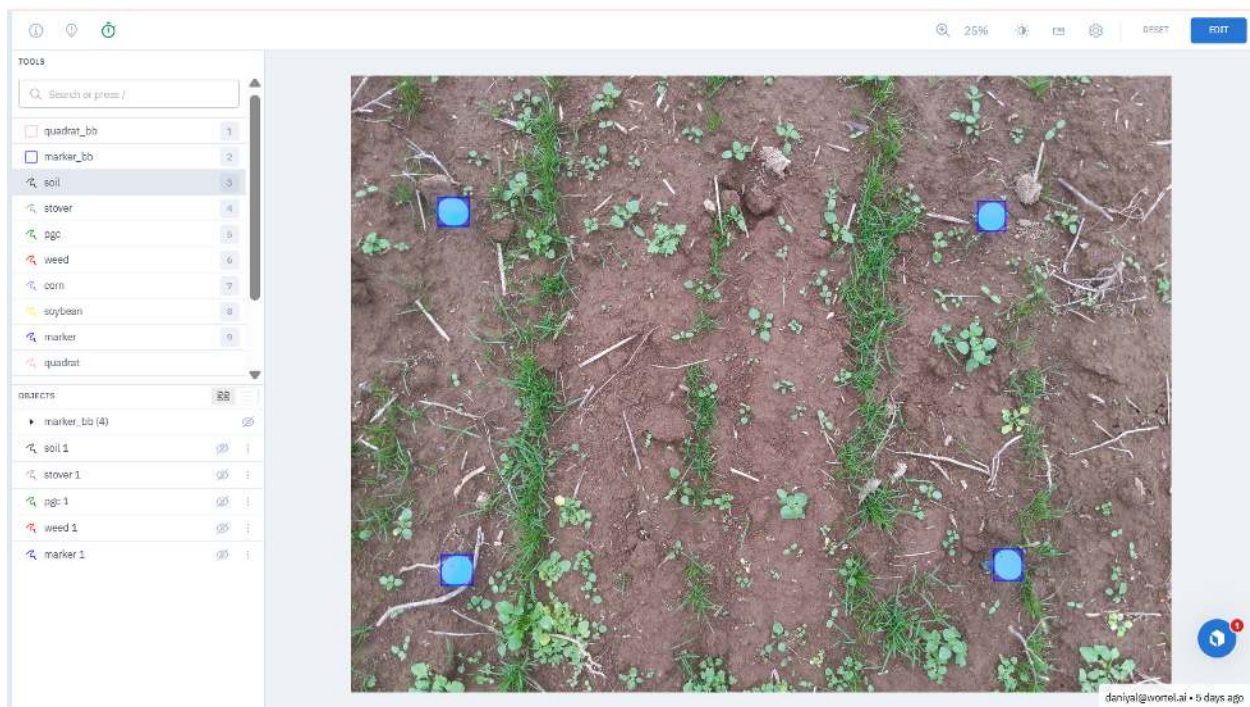
[Figure 5.12] Before Annotations.





[Figure 5.13] Before Annotations.

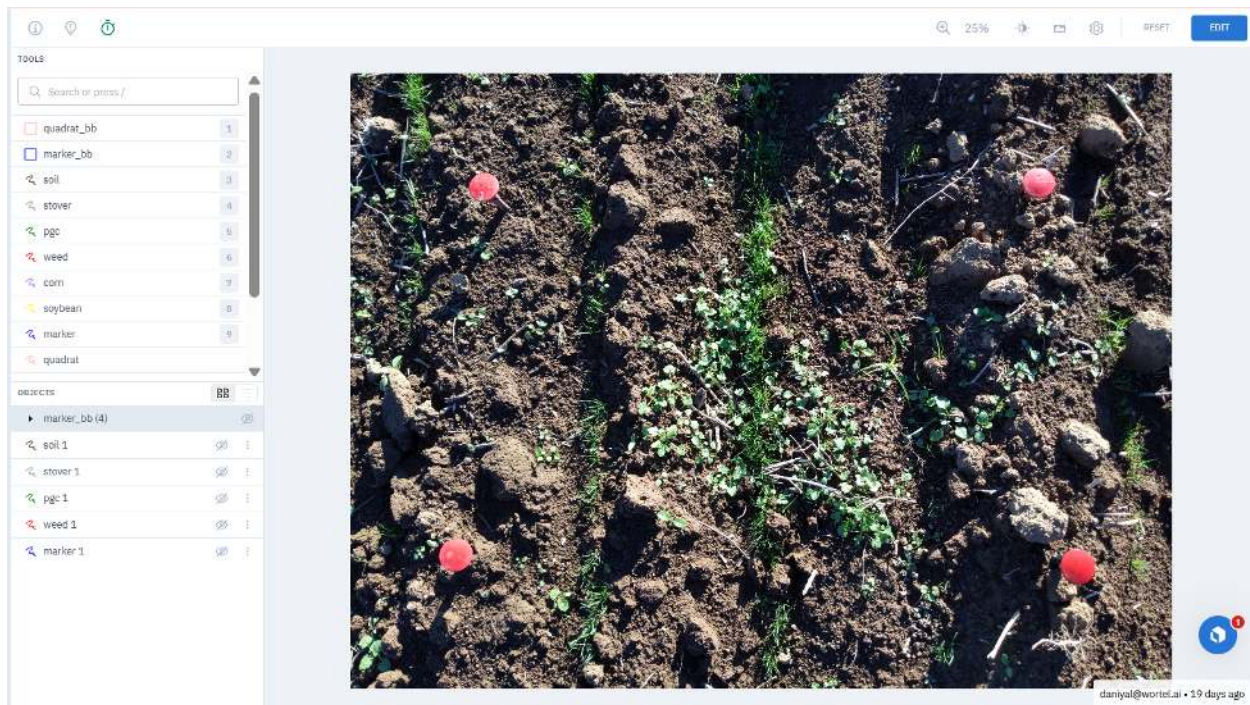




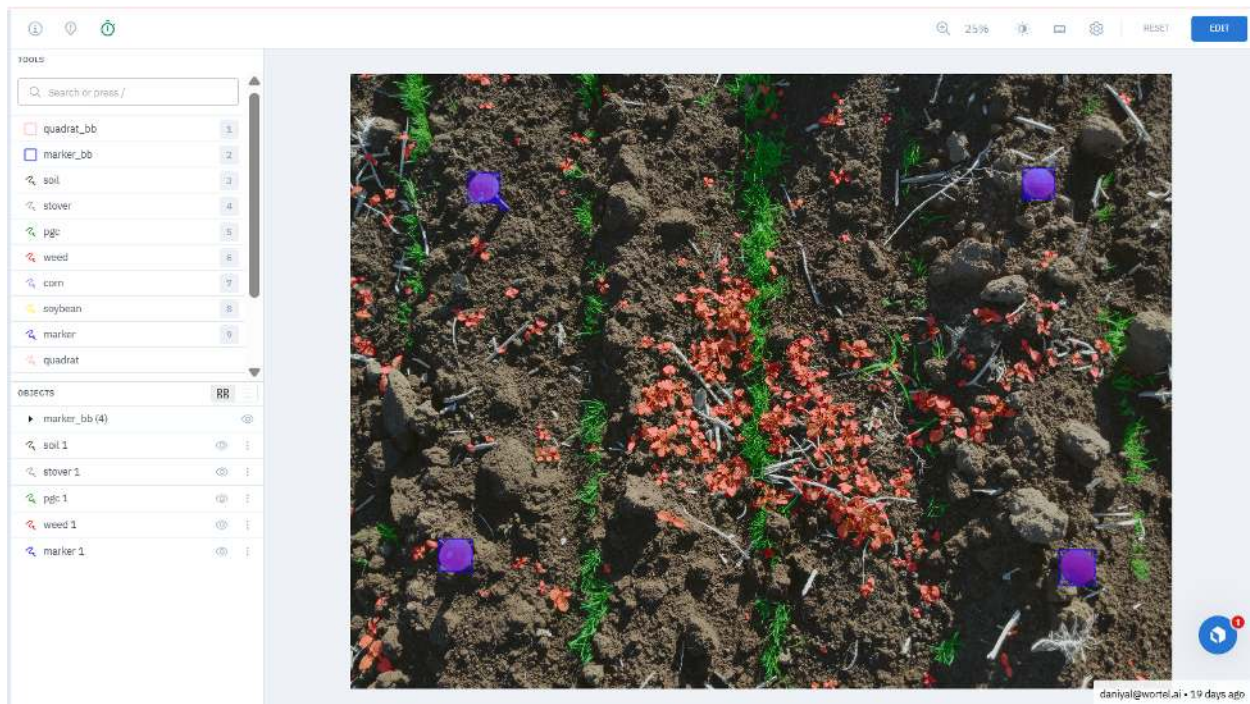
[Figure 5.14] Before Annotations.



[Figure 5.14].



[Figure 5.15] Before Annotations.



Here is the video that shows samples of annotated images from the PGCView_v2 project. Video Link:

<https://www.loom.com/share/acc25d8a71ba48f8873edf71b3df7a75?sid=9462f098-c383-4e7c-b781-676ad1b5698>

6. Conclusion

This project successfully delivered a high-quality annotated dataset for **PGCView_v2** using **Labelbox**. Both bounding box and segmentation annotations were created, reviewed, corrected, and finalized with client feedback. The dataset will support future agricultural research and machine learning model development.